(ii) From the Fig.2.5

Base of a parallelogram (b) = 9 cm, Height of a parallelogram (h) = 16 cm.

Area of a parallelogram = $b \times h$ $sq.units$ Therefore, Area = 9×16 = 144 $sq.\ cm$

Thus, area of the parallelogram is 144 $sq.\ cm$.

Perimeter of the parallelogram = sum of the length of the four sides.

= (18+9+18+9) = 54 cm.

Example 2.2 One of the sides and the corresponding height of the parallelogram are 12 m and 8 m respectively. Find the area of the parallelogram.

Solution

Given: b = 12 m, h = 8 m

Area of the parallelogram = $b \times h$ $sq.units$

= 12 × 8 = 96 $sq.m$

Therefore, Area of the parallelogram = 96 $sq.m$.

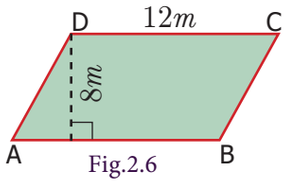


Fig.2.6

Example 2.3

Find the height 'h' of the parallelogram whose area and base are 368 $sq.\ cm$ and 23 cm respectively.

Solution

Given: Area = 368 $sq.\ cm$, base b = 23 cm

Area of the parallelogram = 368 $sq.\ cm$

$b \times h$ = 368

23 × h = 368

$h = \frac{368}{23}$ = 16 cm

Thus, the height of the parallelogram = 16 cm.

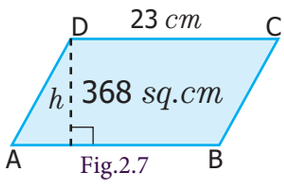


Fig.2.7

Example 2.4

A parallelogram has adjacent sides 12 cm and 9 cm. If the distance between its shorter sides is 8 cm, find the distance between its longer side.

Solution

Given that the adjacent sides of parallelogram are 12 cm and 9 cm

If we choose the shorter side as base, that is b = 9 cm then distance between the shorter sides is height, that is h = 8 cm

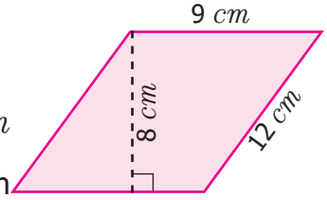


Fig.2.8

Area of parallelogram = $b \times h$ $sq.units$ = 9 × 8 = 72 $sq.\ cm$.

Again, if we choose longer side as base, that is b = 12 cm then distance between longer sides is height. Let it be 'h' units.

7th_Maths_T1_EM_Chp2.indd 37 10/12/2022 17:49:21